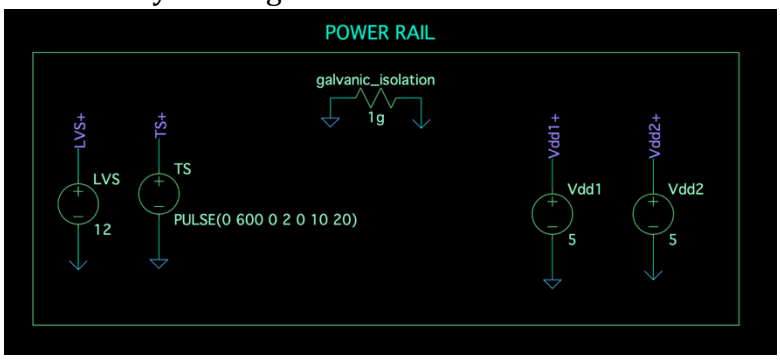
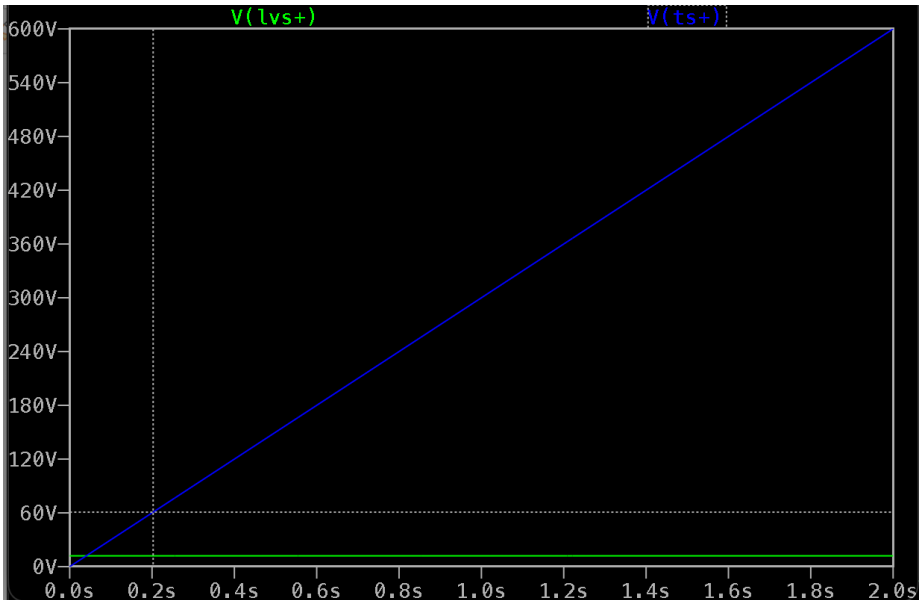
Red Light System Simulation

1. LAYING THE FUNDAMENTALS:

- The Tractive System (TS) here is assumed to represent the voltage across the DC-Link Capacitor for simplicity. The TS is represented by a voltage source that varies from 0 to 600V in 2 seconds. (ramp signal)
- The threshold voltage is assumed to be 60V. The TS ramp signal crosses 60V at 0.2 seconds.
- The LVS system is represented by a 12V voltage source.
- Galvanic Isolation is achieved in simulation by separating the TS_GND (triangle) and LVS_GND (arrow) using 1GΩ resistor.
- Additional 5V voltage sources (Vdd1, Vdd2) are introduced to power the ICs.

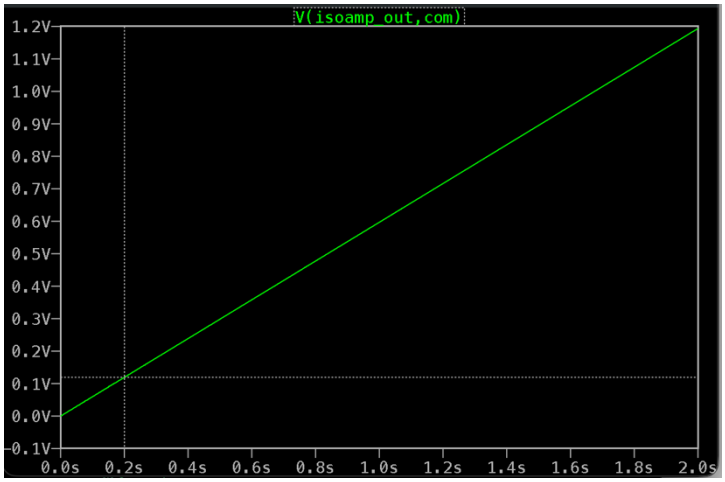




The blue line represents the TS ramp signal (0 – 600V). It crosses 60V (threshold) at 0.2s.
The green line represents LVS and is at a constant DC value of 12V.

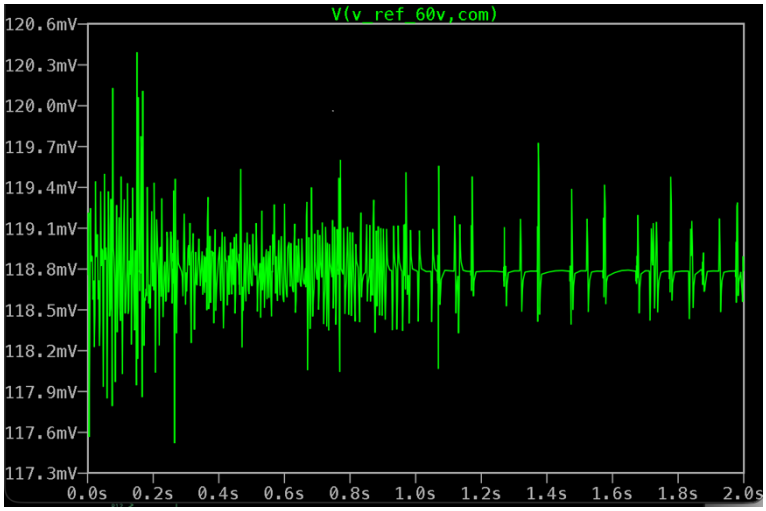
2. READING THE TS VOLTAGE:

- The output of the Isolation amplifier (via. Voltage-dependent voltage source) is 119mV at 0.2 seconds. (119mV represents 60V of TS voltage)
- At the end of 2s, it reaches 1.189V which represents 600V of TS Voltage.
- See: 'docs/TS_Voltage_Reader.md' for schematic.



3. GENERATING 60V THRESHOLD REFERENCE:

- The TL431 shunt regulator maintains 2.5V at its cathode upon using the configured schematic.
- A voltage divider is employed to achieve a constant DC value of 119mV (represents 60V of TS)
- See: 'docs/Generating_reference_voltage.md' for schematic.

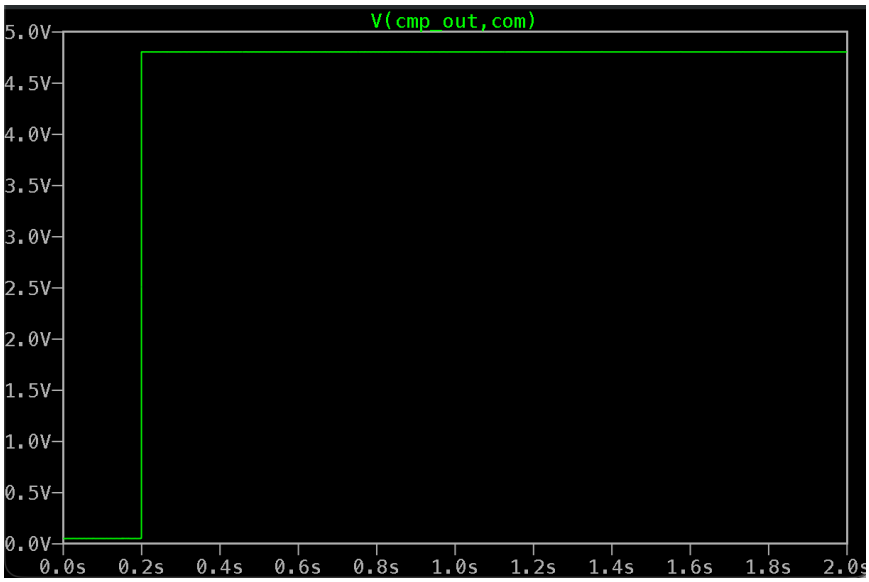


The Spikes occur due to the action of Isolation amplifier and the regulation action of TL431 itself. They are negligible (p-p value: 3mV)

The average value: ~ 119mV

4. COMPARATOR

- The comparator output is supposed to go high when the read TS voltage exceeds the threshold voltage according to the schematic. (threshold cross-over occurs at 0.2 seconds)
- The output is pulled up to 5V using pullup resistor and clamping action by Zener diode.
- See: 'docs/comparator.md' for schematic.

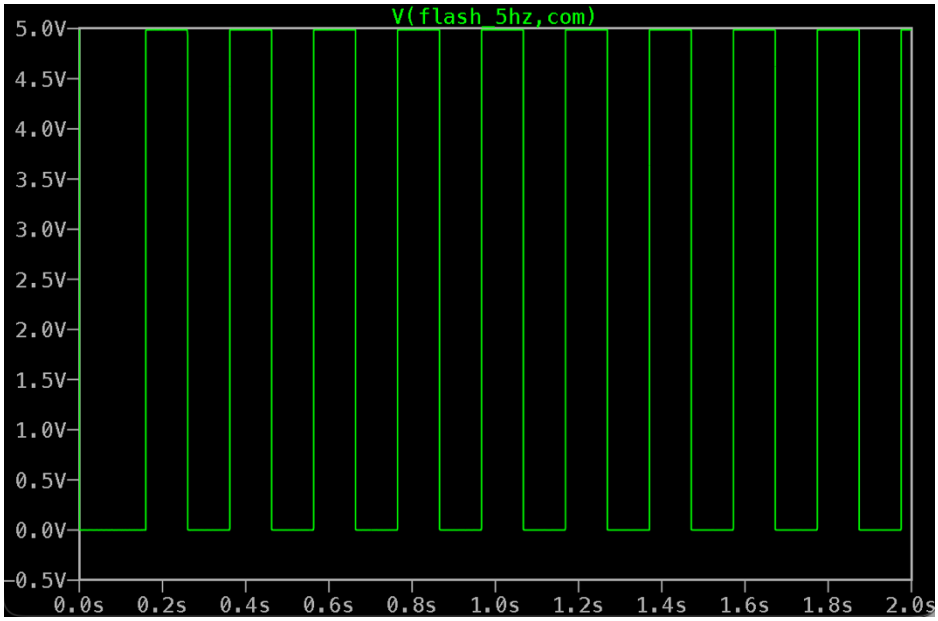


At 0.2 seconds, the TS voltage crosses 60V

The comparator goes high at that exact instant.

5. GENERATING 5Hz, 50% DT, SQUARE WAVE

- See: ‘docs/timer.md’ for working and schematic.

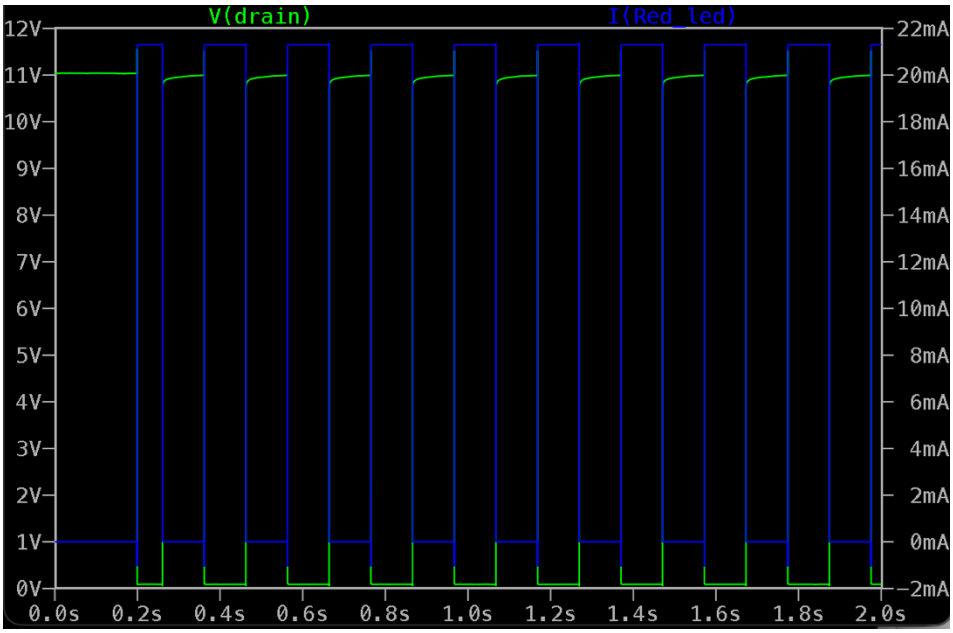


A square wave output with a time period of 200ms (f = 5 Hz).

Peak value: 5V

6. RED LIGHT CIRCUIT

- The AND gate goes output high when the square wave is positive and the comparator is high.
- Its output is fed to the gate of NMOS, which allows the LED to turn ON when the gate is high.
- See: ‘docs/red_light_circuit.md’



The current through the LED represents its ON state. (Blue line)

At exactly 0.2s, current is allowed to pass through the LED. (21mA)

The current waveform is a square wave of 5Hz, meaning, the LED is blinking at 5Hz.

Before 0.2s, the LED stays OFF (current = 0)